Kinect Lite: Explanation of Results

What was tested

Kinect depth images contain **spikes** (pixels with a wildly wrong distance) and **holes** (pixels with no distance). Two cleaners were compared: our **9×9 median** and the **Fu et al. (2012) paper pipeline**. Each cleaned sequence was fed to two trackers, **depth-only ICP** and **RGB-D odometry** (colour + depth), and the estimated camera path was compared with the true path. ATE is total drift (cm); RPE is per-step error (mm); lower is better.

1. Depth accuracy (ICL, against true depth)

Method	MAE (mm)	RMSE (mm)	within 1 cm
midrule No cleaning	39.2	187.5	42.2% Median 9×9 (ours)
18.8	**55.4**	**44.7%** Paper (Fu et al.)	37.7
186.0	33.7%		
bottomrule			

- The raw RMSE (187 mm) is almost 5× its MAE, so **spikes dominate the error**.
- The median votes spikes out: the error roughly halves and RMSE drops to 55 mm.
- The paper's bilateral filter weights neighbours by depth similarity, so a spike only averages with itself and survives. Its error barely changes.

2. Camera tracking (ATE in cm)

Method	ICL ICP	ICL RGB-D	TUM ICP	TUM RGB-D
midrule True depth	0.19	0.81	–	– No cleaning
17.07	23.59	13.97	2.14 Median (ours)	14.94
0.98	13.32	**2.08** Paper	**0.95**	26.0
3.31	2.10			
bottomrule				

The best cleaner depends on the tracker:

- **Depth-only ICP favours the paper.** ICP discards far-off matches, so leftover spikes cost little, but it relies on surface normals. The paper's smooth surfaces give good normals; the median leaves a striped error pattern that biases each step slightly and accumulates as drift.
- **RGB-D odometry favours the median.** It projects every pixel using its depth, so each surviving spike drags the alignment. With the paper's output 49 of 149 steps were off by more than 20 mm; with the median, none. The median's 0.98 cm is close to the 0.81 cm from perfect depth.
- **TUM RGB-D is a tie** (2.08–2.14 cm): real sensor noise has fewer spikes than ICL's simulated noise, so cleaning matters less.

- The paper's ambiguous hole-prediction step does not matter: all three readings agree within 0.06 cm, since holes are only about 1% of pixels.

3. Should holes be filled?

Fill method	Fill err (mm)	ICL RGB-D	TUM RGB-D	
midrule None (keep holes)	–	**0.98**	**2.08**	0
87.0	1.44	3.13	210 ms Navier–Stokes	
1.46	3.89	224 ms Nearest pixel	22.3	
4.16	50 ms Region growing (small)	15.5	1.01	
52 ms Temporal (paper)	**12.2**	1.24	2.51	
bottomrule				

No fill improves RGB-D odometry. A filled pixel is a guess, and the tracker does better ignoring a missing pixel than using a wrong one. Telea, the obvious choice, is among the least accurate because it blends the near and far sides of a depth edge. The fills that change the least hurt the least.

Why the best cleaner flips with the tracker

The depth image has two kinds of mistakes:

- **Spikes:** a few pixels with a very wrong distance.
- **Roughness:** surfaces are slightly bumpy instead of flat, even where the wall is flat.

Each cleaner fixes only one of them: the **median removes spikes** but leaves the roughness; the **paper smooths surfaces** but leaves the spikes.

Each tracker is bothered by only one of them: **depth-only ICP** lines up surfaces, so it cares about roughness and mostly ignores spikes; **RGB-D odometry** uses depth at every pixel, so it cares about spikes, while the colour image makes it less sensitive to roughness.

	Median (fixes spikes)	Paper (fixes roughness)
Depth-only ICP (bothered by roughness)	still drifts: 14.94 cm	works: 0.95 cm
RGB-D odometry (bothered by spikes)	works: 0.98 cm	fails: 26.0 cm

So the median fixes the problem RGB-D has, and the paper fixes the problem ICP has. *Caveat:* the "roughness hurts ICP" part is the project's explanation, not directly proven by the experiments. The numbers only show that the paper helps ICP a lot.

Conclusion

System uses	Use	Why
midrule RGB-D odometry	9×9 median, no hole filling	Error halved, best tracking, under 1 min for 450 frames Depth-only ICP
Paper pipeline	4–16× less drift, but about 26 min for 450 frames Display only	Median + Navier–Stokes or nearest
About 4× more accurate than Telea; do not feed to the tracker		
bottomrule		